ANSWER KEY – World plotting grid (Unit 4 Lesson 6) ● earthquake ▲ volcano

	180-150 W	150-120 W	120-90 W	90-60 W	60-30 W	30-0 W	0-30 E	30-60 E	60-90 E	90-120 E	120-150 E	150-180 E
60-90 N		● 3				▲ 18						
30-60 N		● 1 ▲ 11					▲ 15	● 8			● 6 ● 10 ▲ 12	
0-30 N	▲ 16		● 9 ▲ 19	● 5					● 7	● 4	▲ 13	
0-30 S				▲ 17			▲ 20			▲ 14		
30-60 S				● 2								
60-90 S												

Shaded boxes contain data points. Points sit in the center of their box (the grid is 30° x 30°, so exact position doesn't matter). #13 Mount Pinatubo is at exactly 120° E, so it goes in the 120-150 E box.

Pattern: points line up along plate edges (Pacific 'Ring of Fire': 1,2,3,6,9,10,11,12,13,14,17,19). #16 Mauna Loa (hot spot) is the exception.

Teacher key · Data table: grid box and boundary type

#	Type	Place	Lat., long.	Row	Column	Boundary
1	quake	San Francisco, USA	38 N, 123 W	30-60 N	150-120 W	Transform (San Andreas)
2	quake	Valdivia, Chile	38 S, 73 W	30-60 S	90-60 W	Convergent (subduction)
3	quake	Anchorage, Alaska	61 N, 148 W	60-90 N	150-120 W	Convergent
4	quake	Sumatra, Indonesia	3 N, 96 E	0-30 N	90-120 E	Convergent
5	quake	Port-au-Prince, Haiti	18 N, 72 W	0-30 N	90-60 W	Transform
6	quake	Tohoku, Japan	38 N, 142 E	30-60 N	120-150 E	Convergent
7	quake	Nepal	28 N, 85 E	0-30 N	60-90 E	Convergent (continental collision)
8	quake	Kahramanmaras, Turkey	37 N, 37 E	30-60 N	30-60 E	Transform
9	quake	Michoacan, Mexico	18 N, 102 W	0-30 N	120-90 W	Convergent
10	quake	Kobe, Japan	34 N, 135 E	30-60 N	120-150 E	Crustal fault in a convergent zone
11	volcano	Mount St. Helens, USA	46 N, 122 W	30-60 N	150-120 W	Convergent
12	volcano	Mount Fuji, Japan	35 N, 139 E	30-60 N	120-150 E	Convergent
13	volcano	Mount Pinatubo, Philippines	15 N, 120 E	0-30 N	120-150 E	Convergent
14	volcano	Krakatau, Indonesia	6 S, 105 E	0-30 S	90-120 E	Convergent
15	volcano	Mount Etna, Italy	38 N, 15 E	30-60 N	0-30 E	Convergent region (complex)
16	volcano	Mauna Loa, Hawaii	19 N, 156 W	0-30 N	180-150 W	None: hot spot inside a plate
17	volcano	Cotopaxi, Ecuador	1 S, 78 W	0-30 S	90-60 W	Convergent
18	volcano	Eyjafjallajokull, Iceland	64 N, 20 W	60-90 N	30-0 W	Divergent (Mid-Atlantic Ridge)
19	volcano	Popocatepetl, Mexico	19 N, 99 W	0-30 N	120-90 W	Convergent
20	volcano	Nyiragongo, DR Congo	2 S, 29 E	0-30 S	0-30 E	Divergent (East African Rift)

Pattern: most earthquakes and volcanoes happen along plate edges (Pacific Ring of Fire, Alpine-Himalayan belt, Caribbean). Exception to discuss: Mauna Loa (hot spot).

#13 Pinatubo is at exactly 120 E: use the 120-150 E box. Grid boxes are 30 × 30 degrees, so the pattern matters more than exact position.